

MACHINE LEARNING - Set 1: Scenario Based Questions

Topic: Identify Problem Type & Outline Logic

1. A real estate company wants to develop a system that predicts house prices based on square footage, number of bedrooms, and location.

Type: Supervised – Regression

Logic:

1. Collect data: sq ft, bedrooms, location, price
2. Preprocessing: Handle null values, One-Hot Encode location
3. Split: Train/Test 80/20
4. Model: Linear Regression / XGBoost Regressor
5. Evaluate: RMSE, MAE, R2 Score
6. Predict price for new house

2. A bank wants to build a model to detect fraudulent transactions by analyzing customer spending behavior and transaction history.

Type: Supervised - Classification (Anomaly/Imbalanced)

Logic:

1. Collect spending behavior, transaction history
2. Preprocess - Handle imbalance (SMOTE), scaling
3. Model - Logistic Regression / Random Forest / Isolation Forest
4. Evaluate - Precision, Recall, F1 (Accuracy vended)
5. Deploy with alert system

3. A supermarket wants to segment its customers based on their shopping patterns to provide personalized promotions.

Type: Unsupervised - Clustering

Logic:

1. Collect shopping patterns - frequency, amount, category
2. Preprocess – Scaling
3. Use Elbow method to find K
4. Model - K-Means Clustering Analyze clusters & give personalized offers

4. A company wants to estimate an employee's salary based on their years of experience, job title, and education level.

Type: Supervised - Regression

Logic:

1. collect Features: exp, job title, education.
2. Model: Linear / Ridge Regression.
3. Metric: RMSE

5. An email provider wants to automatically classify incoming emails as spam or not spam based on their content and sender details.

Type: Supervised - Classification (Text Classification)

Logic:

Collect emails with label Spam/Not SpamText

Preprocess - tokenization, stopwords, TF-IDF / Word2VecModel - Naive Bayes / SVM

Evaluate- Accuracy, F1-Score

6. A business wants to analyze customer reviews of its products and determine whether the sentiment is positive or negative.

Type: Supervised - Classification (NLP - Sentiment Analysis)

Logic:

1. Collect reviews
2. Text Preprocess + Sentiment labeling (Positive/Negative)
3. Model - Logistic Regression / BERT
4. Evaluate - Confusion Matrix, F1

7. An insurance company wants to predict whether a customer is likely to file a claim in the next year based on their driving history and demographics.

Type: Supervised - Classification (Binary)

Logic:

1. Data: driving history, demographics, label: claim yes/no
2. Preprocess - encoding, scaling
3. Model - Logistic / Random Forest Classifier
4. Evaluate - AUC-ROC, Recall Predict next year claim probability

8. A streaming platform wants to recommend movies to users by grouping them based on their viewing preferences and watch history.

Type: Unsupervised - Clustering + Recommendation System

Logic:

1. Collect viewing history, preferences
2. Preprocess & create user profile vectors
3. Model - K-Means to group similar users OR Collaborative Filtering
4. Recommend movies liked by similar group

9. A hospital wants to predict the recovery time of patients after surgery based on their age, medical history, and lifestyle habits.

Type: Supervised - Regression

Logic:

- 1.Features: age, medical history, habits. Target: days.
- 2.Model: Regression (Random Forest Regressor).
- 3.Metric: MAE / RMSE

10. A university wants to predict a student's final exam score based on study hours, attendance, and past academic performance.

Type: Supervised - Regression

Logic:

- 1.Features: study hours, attendance, past scores.
- 2.Target: final score (number). Model: Linear Regression.
- 3.Metric: R2 Score, RMSE